Robustness of the weighted BPX factor for $M + \sqrt{\beta}K$

Proof of Assumption 3.4 of the preconditioning paper (draft for checking)

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Status. Steps marked **[proved]** are proved here completely. Steps marked **[cited]** use classical multilevel results, stated precisely as Lemmas 4 and 5, with sources. The argument was checked step by step and numerically (review of 4 Nov. 2026, `proof_bpx_robust_review.md`). Notation: $\nu := \sqrt{\beta} > 0$ (the paper uses ε for the solver accuracy).

1 Setting

Let $\Omega = (0,1)^d$, $h_\ell = 2^{-\ell}$, and let $V_\ell \subset H_0^1(\Omega)$ be the Q_1 finite element space on the uniform grid of mesh size h_ℓ, $\ell = 1, \dots, L$. The spaces are nested, $V_1 \subset \cdots \subset V_L =: V$. For $u \in V_\ell$ let $y \in \mathbb{R}^{n_\ell}$ be its vector of nodal values. Define

$$a(u,v) = (u,v) + \nu(\nabla u, \nabla v), \qquad t_\ell := \nu h_\ell^{-2} = \nu 4^\ell, \qquad \mu_\ell := 1 + t_\ell.$$

The matrix of a in the nodal basis of V_ℓ is $A_\ell = M_\ell + \nu K_\ell$; on the finest level $A_L = A_\beta$ of the paper.

BPX factor. P_ℓ maps nodal values on level ℓ to nodal values on level L of the same function (the natural embedding $V_\ell \subset V_L$; $P_L = I$). $\operatorname{diag}(A_\ell) = \delta_\ell I$ is constant on the uniform grid. With weights $\omega_\ell > 0$,

$$C = \sum_{\ell=1}^{L} \omega_\ell^2 P_\ell P_\ell^T, \qquad \omega_\ell^2 = \frac{\tilde\theta_\ell}{\delta_\ell}.$$

With $\theta_\ell = \min(1, t_\ell)^s$, $s \geq 0$, we use the *normalized* weights (as in the paper, eq. (theta))

$$\tilde\theta_\ell = \frac{\theta_\ell}{\theta_L} = \Big(\frac{\min(1,t_\ell)}{\min(1,t_L)}\Big)^s.$$

Since $\tilde\theta_\ell$ differs from θ_ℓ by the factor θ_L^{-1}, which does not depend on ℓ, C only changes by a positive scalar. So $\kappa(CA_\beta)$ is the same for both choices. The *interval* $[c_0, c_1]$ below, however, is independent of β only for the normalized weights (relevant for the heat problem, Corollary 13).

Claim (Assumption 3.4). For $s > 0$ there are $0 < c_0 \leq c_1$, depending only on d and s, with $\operatorname{spec}(CA_\beta) \subset [c_0, c_1]$ for all $L \geq 1$ and $\nu > 0$.

2 Reduction to a decomposition problem

Lemma 1 (fictitious-space identity; Griebel–Oswald 1995, Xu–Zikatanov 2002)**.** ***[proved]*** *Let $C = \sum_k R_k D_k^{-1} R_k^T$ with D_k symmetric positive definite and $\sum_k \operatorname{range}(R_k) = \mathbb{R}^n$. Then C is SPD and, for every $x \in \mathbb{R}^n$,*

$$x^T C^{-1} x = \min\Big\{ \sum_k y_k^T D_k y_k \;:\; \sum_k R_k y_k = x \Big\}.$$

Proof. C is SPD because $x^TCx = \sum_k |D_k^{-1/2}R_k^Tx|^2 = 0$ forces $R_k^Tx = 0$ for all k, hence $x \perp \sum_k \operatorname{range} R_k = \mathbb{R}^n$. For any decomposition $x = \sum_k R_k y_k$, by Cauchy–Schwarz,

$$x^TC^{-1}x = \sum_k y_k^T R_k^T C^{-1}x \le \Big(\sum_k y_k^T D_k y_k\Big)^{1/2}\Big(\sum_k x^TC^{-1}R_kD_k^{-1}R_k^TC^{-1}x\Big)^{1/2},$$

and the last sum equals $x^TC^{-1}CC^{-1}x = x^TC^{-1}x$. Hence $x^TC^{-1}x \le \sum_k y_k^T D_k y_k$. Equality holds for $y_k = D_k^{-1}R_k^TC^{-1}x$, which is a decomposition of x. □

Apply this with $R_\ell = P_\ell$ and $D_\ell = \omega_\ell^{-2}I = (\delta_\ell/\tilde\theta_\ell)I$; $P_L = I$ makes the ranges span. Let x be the nodal vector of $u \in V$ and y_ℓ that of $u_\ell \in V_\ell$; then $\sum_\ell P_\ell y_\ell = x$ means $\sum_\ell u_\ell = u$, and $x^TA_\beta x = a(u,u)$. So

$$x^TC^{-1}x = \min_{u=\sum_\ell u_\ell,\ u_\ell\in V_\ell} \sum_\ell \frac{\delta_\ell}{\tilde\theta_\ell}|y_\ell|^2. \tag{1}$$

Since $\operatorname{spec}(CA_\beta) \subset [c_0, c_1] \iff c_1^{-1}A_\beta \preceq C^{-1} \preceq c_0^{-1}A_\beta$, the claim is equivalent to

$$c_1^{-1}\,a(u,u) \;\le\; x^TC^{-1}x \;\le\; c_0^{-1}\,a(u,u) \qquad \text{for all } u \in V. \tag{2}$$

3 From nodal coefficients to L^2 norms

Lemma 2. ***[proved]*** *For $v \in V_\ell$ with nodal vector y,*

$$\Big(\frac{h_\ell}{3}\Big)^d|y|^2 < \|v\|^2 < h_\ell^d|y|^2, \qquad \delta_\ell = \Big(\frac{2h_\ell}{3}\Big)^d\big(1+3d\,t_\ell\big) \in \Big[\Big(\frac{2h_\ell}{3}\Big)^d\mu_\ell,\ 3d\Big(\frac{2h_\ell}{3}\Big)^d\mu_\ell\Big].$$

Proof. $\|v\|^2 = y^TM_\ell y$ with $M_\ell = M_1^{\otimes d}$, and the 1D mass matrix $\frac{h}{6}\operatorname{tridiag}(1,4,1)$ has eigenvalues $\frac{h}{6}(4+2\cos\phi_j) \in (\frac{h}{3}, h)$. The diagonal entries are $(M_\ell)_{ii} = (2h/3)^d$ and $(K_\ell)_{ii} = d\cdot\frac{2}{h}(\frac{2h}{3})^{d-1}$, so $\delta_\ell = (\frac{2h}{3})^d + \nu\frac{2d}{h}(\frac{2h}{3})^{d-1} = (\frac{2h}{3})^d(1+3d\nu h^{-2})$. Finally $\mu \le 1+3dt \le 3d\mu$ for $t \ge 0$ and $d \ge 1$. □

Write $\lambda_\ell := \mu_\ell/\tilde\theta_\ell$ and

$$N(u) := \inf_{u=\sum_\ell u_\ell,\ u_\ell\in V_\ell} \sum_\ell \lambda_\ell\|u_\ell\|^2.$$

By Lemma 2, for each term $(\frac{2}{3})^d\lambda_\ell\|u_\ell\|^2 \le \frac{\delta_\ell}{\tilde\theta_\ell}|y_\ell|^2 \le 3d\,2^d\lambda_\ell\|u_\ell\|^2$ (the upper bound uses $h_\ell^d|y_\ell|^2 < 3^d\|u_\ell\|^2$, the lower bound $\|u_\ell\|^2 < h_\ell^d|y_\ell|^2$). Taking the infimum in (1),

$$\big(\tfrac{2}{3}\big)^d N(u) \;\le\; x^TC^{-1}x \;\le\; 3d\,2^d\,N(u). \tag{3}$$

It remains to show $K_1^{-1}a(u,u) \le N(u) \le K_0\,a(u,u)$ with K_0, K_1 independent of L and ν. Then (2) holds with

$$c_0 = \frac{1}{3d\,2^dK_0}, \qquad c_1 = \big(\tfrac{3}{2}\big)^dK_1, \qquad \frac{c_1}{c_0} \le 3d\cdot 3^d\cdot K_0K_1.$$

The factor $3d\cdot 3^d$ comes only from Lemma 2, i.e. from Jacobi scaling and the nodal basis. The comparison (3) itself is nearly sharp, but its extremes are not attained by the same functions as those of $N(u)/a(u,u)$, so the actual $\kappa(CA_\beta)$ is much smaller. In the unresolved limit $\nu \to 0$ one observes $\kappa(CM) \to 3^d/(1-4^{-s})$ (4.000 for $d=1$, 11.93 for $d=2$, $L=6$, both with $s=1$).

4 Two regimes and the weights

Since $t_\ell = \nu 4^\ell$ increases by the factor 4 per level, there are two regimes.

- *Unresolved* ($t_L < 1$): all levels are mass dominated. Then $\tilde{\theta}_\ell = (t_\ell/t_L)^s = 4^{-s(L-\ell)}$ and $\lambda_\ell = (1+t_\ell)\,4^{s(L-\ell)}$.
- *Resolved* ($t_L \geq 1$): let $\ell^* = \min\{\ell : t_\ell \geq 1\}$. Then $\tilde{\theta}_\ell = \theta_\ell$, and $\lambda_\ell = 1 + t_\ell \in [t_\ell, 2t_\ell]$ for $\ell \geq \ell^*$, while $\lambda_\ell = (1+t_\ell)t_\ell^{-s} \in [t_\ell^{-s}, 2t_\ell^{-s}]$ for $\ell < \ell^*$. Moreover $t_{\ell^*} < 4$ if $\ell^* > 1$.

Lemma 3. ***[proved]*** *For $s > 0$, in both regimes,*

$$\textit{(a)}\quad t_\ell \leq \lambda_\ell \ \textit{ for all } \ell, \qquad \textit{(b)}\quad S := \sum_{\ell=1}^{L} \lambda_\ell^{-1} \leq \frac{4}{3} + \frac{1}{1-4^{-s}}.$$

For $s = 0$, (a) still holds, and $\frac{1}{2}m \leq S \leq m + \frac{4}{3}$, where $m = \#\{\ell : t_\ell < 1\}$ is the number of mass-dominated levels: $m = L$ in the unresolved regime and $m = \ell^* - 1 = \lceil \log_4(1/\nu) \rceil - 1$ in the resolved one.

Proof. (a) If $t_\ell \geq 1$ then $\lambda_\ell = 1 + t_\ell > t_\ell$. If $t_\ell < 1$ then $\lambda_\ell \geq 1 > t_\ell$, since $\tilde{\theta}_\ell \leq 1$ in both regimes. (b) Unresolved: $\lambda_\ell^{-1} \leq 4^{-s(L-\ell)}$, so $S \leq \sum_{j\geq 0} 4^{-sj} = 1/(1-4^{-s})$. Resolved: for $\ell \geq \ell^*$, $\lambda_\ell^{-1} \leq t_\ell^{-1} = t_{\ell^*}^{-1} 4^{-(\ell-\ell^*)} \leq 4^{-(\ell-\ell^*)}$, which sums to at most $4/3$. For $\ell < \ell^*$, $\lambda_\ell^{-1} \leq t_\ell^s = t_{\ell^*-1}^s 4^{-s(\ell^*-1-\ell)} < 4^{-s(\ell^*-1-\ell)}$, which sums to at most $1/(1-4^{-s})$. For $s = 0$ the bounds $\lambda_\ell^{-1} \leq 1$ on the mass-dominated levels are attained up to the factor $1 + t_\ell < 2$. □

The proof uses only $\tilde{\theta}_\ell \leq 1$ and the geometric decay, so it holds for every $s > 0$. Letting $s \to \infty$ ($\tilde{\theta}_\ell = 0$ for $t_\ell < 1$ when $t_L \geq 1$, and only level L when $t_L < 1$) gives $S \leq 4/3+1$, so the truncated method of Griebel and Oswald is covered as well. Numerically, $\max S = 1/(1-4^{-s})$; the stated bound is valid but not sharp.

5 Two classical ingredients

Let $Q_\ell : L^2(\Omega) \to V_\ell$ be the L^2-orthogonal projection, $Q_0 := 0$.

Lemma 4 (multilevel norm equivalence, Jackson direction)**.** ***[cited]*** *There is C_Q, depending only on d, such that for all L and all $u \in V_L$*

$$\sum_{\ell=1}^{L} h_\ell^{-2} \|(Q_\ell - Q_{\ell-1})u\|^2 \leq C_Q\, |u|_{H^1}^2.$$

Sources:

- Griebel–Oswald, Adv. Comput. Math. 4 (1995) 171–206, proof of Thm. 1, eqs. (23)–(24): Q_1 on $(0,1)^d$ with Dirichlet conditions;
- Bornemann–Yserentant, Numer. Math. 64 (1993) 455–476, Thm. 7.5 (stated for P_1; the proof uses only an approximation property and carries over to Q_1);
- Oswald, *Multilevel Finite Element Approximation*, 1994;
- Xu, SIAM Rev. 34 (1992), Lemma 6.7.

Dirichlet conditions are needed: for $\ell = 1$ the term with $Q_0 = 0$ uses Poincaré's inequality. Numerically $C_Q \approx 0.444$ ($d = 1$, converged) and $C_Q < 0.5$ ($d = 2$). Note that the easy argument via $\|(I - Q_\ell)u\| \lesssim h_\ell |u|_1$ only gives $C_Q = O(L)$; the L-independent bound is the deep part of BPX theory.

Lemma 5 (strengthened Cauchy–Schwarz and inverse inequality)**.** ***[cited]*** *There are C_S and C_I, depending only on d, such that for $k \le \ell$, $u \in V_k$, $v \in V_\ell$,*

$$|(\nabla u, \nabla v)| \le C_S\, 2^{-(\ell-k)/2}\, |u|_{H^1}\, h_\ell^{-1} \|v\|, \qquad |v|_{H^1}^2 \le C_I h_\ell^{-2} \|v\|^2.$$

Sources: Xu, SIAM Rev. 34 (1992), Lemma 6.1, with rate $\gamma = 2^{-1/2}$ for dyadic refinement (after Yserentant, Numer. Math. 49, 1986); restated in Kolev–Xu–Zhu, J. Sci. Comput. 67 (2016), Lemma 4.3. The proof is for P_1. It transfers to Q_1 because every d-linear function is harmonic on each coarse element (integration by parts on coarse elements, a polynomial trace-inverse estimate, and a boundary-strip estimate for v); this transfer is our argument. Numerically $C_S \approx 2.63$ ($d = 1$) and 3.46 ($d = 2$), and the rate $1/2$ is sharp in 1D.

For Q_1 on a uniform grid the inverse inequality holds with the sharp constant $C_I = 12d$: in 1D the generalized eigenvalues of (K_1, M_1) are $\frac{12}{h^2}\frac{\sin^2(\phi/2)}{2+\cos\phi} < \frac{12}{h^2}$, and in d dimensions they are sums of d such values. Hence $C_H = 12d + 2C_S\sqrt{12d}/(\sqrt{2}-1)$.

Lemma 6 (H^1 part of the upper bound)**.** ***[proved from Lemma 5]*** *For every decomposition $u = \sum_\ell u_\ell$, $u_\ell \in V_\ell$,*

$$\nu |u|_{H^1}^2 \le C_H \sum_\ell t_\ell \|u_\ell\|^2, \qquad C_H := C_I + \frac{2C_S C_I^{1/2}}{\sqrt{2}-1}.$$

Proof. Let $a_\ell := t_\ell^{1/2}\|u_\ell\| = \nu^{1/2} h_\ell^{-1}\|u_\ell\|$. Expanding, $\nu|u|_1^2 = \nu\sum_\ell |u_\ell|_1^2 + 2\nu\sum_{k<\ell}(\nabla u_k, \nabla u_\ell)$. The diagonal is at most $C_I\sum_\ell a_\ell^2$ by the inverse inequality. For $k < \ell$, Lemma 5 and the inverse inequality for u_k give $\nu|(\nabla u_k, \nabla u_\ell)| \le C_S C_I^{1/2} 2^{-(\ell-k)/2} a_k a_\ell$. With $q = 2^{-1/2}$, $\sum_{k<\ell} q^{\ell-k} a_k a_\ell \le \big(\sum_{j\ge 1} q^j\big)\sum_\ell a_\ell^2 = \frac{1}{\sqrt{2}-1}\sum_\ell a_\ell^2$ by the Schur test ($2a_k a_\ell \le a_k^2 + a_\ell^2$). □

6 The two bounds

Lemma 7 (upper bound for a)**.** ***[proved, given Lemma 5]*** *For $s > 0$ and every decomposition $u = \sum_\ell u_\ell$,*

$$a(u,u) \le K_1 \sum_\ell \lambda_\ell \|u_\ell\|^2, \qquad K_1 := \frac{4}{3} + \frac{1}{1-4^{-s}} + C_H.$$

Consequently $a(u,u) \le K_1 N(u)$.

Proof. Mass part: by Cauchy–Schwarz, $\|\sum_\ell u_\ell\| \le \sum_\ell \lambda_\ell^{-1/2}\cdot\lambda_\ell^{1/2}\|u_\ell\| \le S^{1/2}(\sum_\ell \lambda_\ell\|u_\ell\|^2)^{1/2}$, and S is bounded by Lemma 3(b). Stiffness part: Lemma 6 and $t_\ell \le \lambda_\ell$ (Lemma 3(a)). □

Lemma 8 (stable decomposition)**.** ***[proved, given Lemma 4]*** *For every $u \in V$, $N(u) \le K_0\, a(u,u)$ with $K_0 := \max(5, 2C_Q)$, for every $s \ge 0$.*

Proof. Unresolved regime. Take $u_L = u$ and $u_\ell = 0$ otherwise. Then $\lambda_L = 1 + t_L < 2$ and $N(u) \le 2\|u\|^2 \le 2a(u,u)$.

Resolved regime. Take $u_\ell = 0$ for $\ell < \ell^*$, $u_{\ell^*} = Q_{\ell^*}u$, and $u_\ell = (Q_\ell - Q_{\ell-1})u$ for $\ell > \ell^*$. These sum to u, since $Q_L u = u$. For $\ell > \ell^*$, $\lambda_\ell \le 2t_\ell$. For ℓ^*:

- if $\ell^* > 1$: $\lambda_{\ell^*} = 1 + t_{\ell^*} < 5$ and $\|Q_{\ell^*}u\| \le \|u\|$, so this term is at most $5\|u\|^2$;
- if $\ell^* = 1$: $\lambda_1\|Q_1 u\|^2 \le \|u\|^2 + t_1\|(Q_1 - Q_0)u\|^2$.

In both cases

$$N(u) \le 5\|u\|^2 + 2\nu\sum_{\ell\ge 1} h_\ell^{-2}\|(Q_\ell - Q_{\ell-1})u\|^2 \le 5\|u\|^2 + 2C_Q\,\nu|u|_1^2 \le K_0\, a(u,u)$$

by Lemma 4. □

7 Result

Theorem 9. *Let $s > 0$ and use the normalized weights $\tilde{\theta}_\ell$. For all $L \geq 1$ and $\beta > 0$,*

$$\operatorname{spec}(CA_\beta) \subset [c_0, c_1], \qquad c_0 = \frac{1}{3d\, 2^d \max(5, 2C_Q)}, \qquad c_1 = \left(\tfrac{3}{2}\right)^d \Big(\frac{4}{3} + \frac{1}{1-4^{-s}} + C_H\Big).$$

In particular $\kappa(CA_\beta) \leq 3d \cdot 3^d \max(5, 2C_Q)\big(\frac{4}{3} + \frac{1}{1-4^{-s}} + C_H\big)$, independently of h and β. For the unnormalized weights θ_ℓ, $\operatorname{spec}(CA_\beta) \subset \theta_L[c_0, c_1]$ with $\theta_L = \min(1, t_L)^s$: the condition number bound is the same, but c_0 degrades by t_L^s in the unresolved regime. Since the cost bounds of the paper involve c_0 itself, the paper uses the normalized weights.

Proof. Combine (3) with Lemmas 7 and 8. The last sentence follows from the scale invariance of κ. □

Remark 10 ($s = 0$, plain BPX). Lemma 8 still holds, but in Lemma 7 the mass part contributes S, which is of the order of the number of mass-dominated levels. The resulting bound grows like $\log(1/\beta)$ in the resolved regime and like L in the unresolved regime. This matches the observed degradation of plain BPX, and the growth is genuine: for unweighted levels $\lambda_{\max}(CA) \gtrsim m$ (test $u \in V_1$ split equally over the m mass-dominated levels; cf. Kolev–Xu–Zhu, J. Sci. Comput. 67 (2016), Remark 4.5). Yang–Yu–Zhang (arXiv:2610.06987, Thm. 2.1) use $\omega_\ell^2 = (h_\ell^{-2} + c)^{-1}$ with exact M_ℓ^{-1}, i.e. our $s = 0$ with exact mass inverse. Their bound $12dL^2$ loses one factor L at each end through per-level estimates; with Lemmas 4 and 5 it improves to $O(\min(L, \log(1/\nu)) + C_H)$.

Remark 11 (sharpness). The constants are far from sharp: for $d = 3$ the factor $3d \cdot 3^d = 243$ alone exceeds the observed $\kappa(CA_\beta) \leq 20.5$. With the computed C_Q, C_S the bound evaluates to about $2.6 \cdot 10^3$ ($d = 1$) and $2.9 \cdot 10^4$ ($d = 2$, $C_I = 12d$), against observed maxima 6.14 and 11.5; the actual $K_0 \leq 1.28$ and $K_1 \leq 29.5$. The proof explains the structure (geometric damping makes the mass part summable; the floor comes from Jacobi scaling), not the size of the constants.

Remark 12 (relation to the literature). Robust additive multilevel preconditioning of $-\varepsilon\Delta + I$ is known: Griebel–Oswald, Adv. Comput. Math. 4 (1995), Thms. 1–2 (nodal BPX and multilevel diagonal scaling for Q_1 with mass-dominated levels dropped, our $s \to \infty$); Bramble–Pasciak–Vassilevski, Math. Comp. 69 (2000) 463–480 (weights $(\varepsilon h_\ell^{-2} + 1)^{-1}$ on L^2-projection differences); Bacuta–Hayes–Jacavage, arXiv:2104.11337 (nodal version with lumped mass, whose level weights are within level-independent factors of ours with $s = 1$ in the resolved regime). What is new here is the Jacobi-scaled factored form, the family $s > 0$, the normalization that makes $[c_0, c_1]$ independent of β, and the explicit constants.

Corollary 13 (time-periodic heat problem)**.** *For frequency k the paper's operator is $A_k = r_k M + \nu K = r_k(M + \nu_k K)$ with $\nu_k = \nu / r_k$. Using in R_k the diagonal of A_k and the normalized weights for ν_k, we get $C_k A_k = C(\nu_k) A_{\nu_k}$ with $C(\nu_k)$ the factor of Theorem 9 for parameter ν_k. Hence* $\operatorname{spec}(C_k A_k) \subset [c_0, c_1]$ *with the* same *c_0, c_1 for all k. This is the hypothesis of the paper's heat theorem.*

Proof. $\operatorname{diag}(A_{k,\ell}) = r_k \operatorname{diag}(M_\ell + \nu_k K_\ell)$, so $\omega_{\ell,k}^2 = r_k^{-1} \omega_\ell(\nu_k)^2$ and $C_k = r_k^{-1} C(\nu_k)$. Then $C_k A_k = C(\nu_k)(M + \nu_k K)$, and Theorem 9 applies with ν_k; its interval does not depend on ν. □